1 Sep 2025

51.504 Machine Learning (Graduate)
ISTD Pillar, September 2025

Lecturer	**: Soh De Wen (dewen_soh@sutd.edu.sg)**
Schedule	**: Tuesdays, 1900 - 2200**
Classroom	**: Physical (Think Tank 9-10, 1.415/1.416)/Online**
Credits	: **12**
TA	**: Ganesh Subramanian (ganesh_subramanian@sutd.edu.sg)**

Subject Description

This graduate level course develops a foundation for research on intelligent data processing. The topics covered include classification, regression, clustering, sequence modelling, recommender problems, generative and discriminative models, model selection and generalization issues, transfer learning, scalability issues, knowledge representations, and various applications.

Learning Objectives

1. Supervised and Unsupervised Learning: Classification, Regression, Clustering
2. Linear and non-linear classification, Recommender problems, Generative modeling
3. Mixture Models, Understanding Generalization, Generative modeling of sequences
4. Graphical Models, Bayesian Networks, Control/decision problems, Uncertainty

Measurable Outcomes

By the end of the course, the students will be able to:

1. List useful real-world applications of machine learning.
2. Implement and apply machine learning algorithms.
3. Choose appropriate algorithms for a variety of problems.

Pre-Requisite/ Co-requisite/ Mutually Exclusive Subject(s)

Calculus, Linear Algebra, Multivariate Calculus, Elementary Probability.

1 Sep 2025

Required or Recommended Text and Readings

Required	
Wasserman (2004), All of Statistics: A Concise Course in Statistical Inference, Springer.	Covers many important topics, and gives computational examples in R.
Bishop (2006): Pattern Recognition and Machine Learning, Springer.	We will mainly be using this book.

Assessment Methods

Components	Weightage	Week	Remarks
Final Exam	30%		Physical, double sided A4 cheatsheet.
Midterm Exam	30%		
Project	20%		More information in coming weeks.
Homework	20%		Please feel free to use computational software and to collaborate with others. Describe your tools and credit your collaborators in your solutions.

Detailed Outline of the Subject

Week	Date	Topics	Physical/ Online	Remarks
1	16/09/2025	Introduction, Probability Review	Physical	
2	23/09/2025	Regression	Physical	HW1 handed out
3	30/09/2025	Classification: Logistic and Perceptron	Online	Project proposal guidelines uploaded
4	7/10/2025	Clustering and Lagrange Multipliers	Physical	
5	14/10/2025	Support Vector Machines	Physical	HW1 due, HW2 handed out
6	21/10/2025	Stochastic Processes, Gaussian Distribution, Gaussian Processes	Physical	
7	28/10/2025	RECESS		
8	04/11/2025	MIDTERMS	Physical	
9	11/11/2025	EM Algorithm, Hidden Markov Models	Physical	Project proposal due
10	18/11/2025	Neural Networks, Deep Learning	Physical	HW2 due, HW3 handed out
11	25/11/2025	Graphical Models, Bayesian Networks and Markov Random Fields	Physical	
12	02/12/2025	Project Presentations	Online	
13	09/12/2025	Reinforcement Learning	Physical	HW3 due
14	16/12/2025	FINALS	Physical	